

MEETLIO

Full SaaS Application — Final Production Readiness Report

Generated: Wednesday, August 26, 2026 | Version 1.0.0

1. Executive Summary & Project Overview

Meetlio is a full-stack, enterprise-grade scheduling and appointment management SaaS platform built with Next.js 15, PostgreSQL, Prisma ORM, and Tailwind CSS. It empowers professionals and businesses to accept client bookings, manage weekly availability, automate cancellation and rescheduling lifecycles, deliver RFC 5545 calendar invites (.ics), send 24h/1h email reminders, receive HMAC-signed outbound webhooks, view audit timelines, inspect real-time PostgreSQL analytics, and enforce subscription plan usage limits.

2. Architecture & Technology Stack

- **Framework & Router:** Next.js 15 (App Router, Server Components, API Routes)
- **Database & ORM:** PostgreSQL 16 + Prisma ORM (Type-safe Schema & Client)
- **Styling & UI Components:** Tailwind CSS, Lucide Icons, Custom UI Design System
- **Authentication:** Bcryptjs password hashing + Server-side Session Cookies
- **Timezone Calculations:** Timezone-aware UTC instant scheduling engine
- **Calendar Integrations:** RFC 5545 compliant iCalendar (.ics) export endpoint
- **Email Notifications:** Nodemailer SMTP integration with HTML/Text templates
- **Webhooks & Security:** HMAC SHA-256 signed event delivery + Rate Limiting
- **Billing Infrastructure:** Provider-independent abstraction (FREE, PRO, BUSINESS)

3. Completed Implementation Breakdown (Phases 1 – 9)

- Phase 1: Core Foundation & Auth:** User registration, login, logout, PostgreSQL connection, and session cookies.
- Phase 2: Event Type Management:** CRUD for meeting types with duration, location, buffers, and slug generation.
- Phase 3: Availability System:** Weekly schedules (Mon-Sun) and specific date overrides (Available / Unavailable).
- Phase 4: Scheduling Engine:** Timezone-aware slot generation considering notice, booking window, and buffers.
- Phase 5: Booking Creation:** Transaction-safe atomic bookings with double-booking race condition prevention.
- Phase 6: Booking Lifecycle:** Guest/Host cancellations, rescheduling, secure tokens, .ics invites, & reminders.
- Phase 7: SaaS Hardening & SEO:** User profile settings, password change, public landing pages, and dynamic SEO.
- Phase 8: Search, Webhooks & Audit:** Multi-field search, pagination, HMAC-SHA256 webhooks, and audit logs.
- Phase 9: SaaS Billing & Limits:** Subscription tiers (FREE, PRO, BIZ), server-side usage checks, and pricing UI.

4. Subscription Tier Limits & Usage Enforcement

- **FREE TIER (\$0/mo):** 1 Event Type | 50 Bookings/mo | 1 Webhook Endpoint
- **PRO TIER (\$19/mo):** Unlimited Event Types | 500 Bookings/mo | 10 Webhooks | Analytics Enabled
- **BUSINESS TIER (\$49/mo):** Unlimited Event Types | Unlimited Bookings | Unlimited Webhooks | Dedicated Support

5. Final Production Audit Results

- **Prisma Client Generation:** **[PASS]** — Generated Prisma Client v6.19.3 cleanly
- **Database Schema Sync:** **[PASS]** — PostgreSQL database meetlio in sync
- **TypeScript Compiler Check:** **[PASS]** — 0 Errors (tsc --noEmit)
- **ESLint Code Quality:** **[PASS]** — No warnings or errors (next lint)
- **Next.js Production Build:** **[PASS]** — 39/39 static & dynamic routes compiled
- **Phase 1-8 Regressions:** **[PASS]** — All booking, availability, and auth engines functional
- **Phase 9 SaaS Billing Suite:** **[PASS]** — All 45 automated functional & limit tests passed

6. How to Run Meetlio

1. Open Terminal or PowerShell
2. Navigate to project root: `cd "C:\Users\H&S TECH\gemini\antigravity\scratch\meetlio"`
3. Run development server: `npm run dev`
4. Open browser at: `http://localhost:3000`